Nivid Pathak

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EXPERIENCE

Data Engineer Assistant **Oct 2025 – Present**
University of Colorado Boulder, Parking Services *Boulder, CO*

- Cut reporting turnaround by 34% for department stakeholders by automating the equipment planning intake workflow with a data-driven tool that extracts structured data and routes requests through an approval pipeline.
- Cut parking assignment turnaround by 43% by replacing a manual routing process with a rule-based system in Python and SQL, translating cross-functional requirements from multiple departments into business rules.
- Cut manual billing review time by 41% by designing PostgreSQL schemas for operational billing data with invoice-parsing logic, data validation checks that improved data quality, and role-based access controls.

Graduate Teaching Assistant **Aug 2026 – Present**
University of Colorado Boulder *Boulder, CO*

- Supported 3 graduate Data Science courses in one semester (Data Science as a Field, Ethical Issues in Data Science, Cybersecurity for Data Science) for 3 faculty instructors, by running office hours and grading projects against course rubrics with written feedback on reproducible analysis and ethical data use.

ML Engineer Intern **Dec 2024 – May 2025**
Semitronik Instruments *Gandhinagar, India*

- Reached 97% validation accuracy on automated defect detection across live inspection pipelines by training, statistically validating, and deploying a PyTorch deep learning computer vision model end to end.
- Cut inference latency by 80% (4.0s to 0.8s) by vectorizing and batching the inference pipeline, then converting the model to ONNX and TensorRT for production deployment.

Data Scientist **Jan 2023 – Nov 2024**
iValue Systech Pvt. Ltd. *Ahmedabad, India*

- Cut false-positive security alerts by 30% across ~50K monthly events by building an XGBoost risk model with SHAP explanations of each alert's drivers, and monitored model performance with a quarterly benchmarking harness that caught drift before production.
- Cut downtime incidents by 22% across 3 clients by building a forecasting pipeline with Prophet, Airflow, and Snowflake.
- Cut executive reporting prep from 2 days to 2 hours by building a Power BI dashboard of business metrics for leadership reviews.
- Cut support query resolution time from 2 days to under 10 minutes by building a GenAI support assistant with RAG (LangChain, ChromaDB, GPT-3.5), and caught an 11-point quality drop before deployment with a RAGAS evaluation pipeline.

PROJECTS

JARVIS, Multi-Agent AI Assistant **May 2026 – Present**

- Handled voice and text requests across 4 specialist agents (calendar, weather, news, sports) running in parallel by building a multi-agent assistant in Python on local LLMs through Ollama, with a swappable Anthropic API backend.
- Raised the router's main evaluation metric, macro-F1, from 0.885 to 0.947 and cut routing latency 2.3x (857ms to 372ms) against a Llama 3.1 8B baseline by fine-tuning a 4-bit Llama 3.2 3B with LoRA on MLX.
- Confirmed the latency gain and flagged the accuracy gain as not yet significant ($p = 0.302$) by designing an offline experiment on 94 held-out examples and testing it with McNemar's exact test and Wilson confidence intervals.

Labor Market Wage Analysis and Minimum Wage Causal Study **Jan 2026 – Oct 2026**

- Predicted occupation wages at $R^2 = 0.97$ ($5.2K RMSE) and classified wage quartiles at 90% accuracy by building an ETL pipeline over ~739K BLS records with 79 engineered lag features and training four model families (Random Forest, Decision Tree, K-Means, FP-Growth).
- Measured the impact of large state minimum wage increases, finding 10th-percentile hourly wages rose ~9.4% (Callaway-Sant'Anna ATT 0.090, 95% CI 0.068 to 0.113) by building an observational difference-in-differences study of 26 treated vs 20 never-treated states.
- Showed the food-service job-loss estimate was not robust (-0.034 full sample vs -0.005, $p = 0.62$, pre-COVID) by stress-testing it with event studies, pre-trend Wald tests, division-by-year fixed effects, lagged treatment, and a placebo outcome.
- Confirmed the wage effect with a synthetic control of Arizona's 2017 increase (0.132 log-point gap, permutation $p = 0.048$, rank 1 of 21), and sized the design's limits with a 500-draw simulation power analysis (80% power MDE of 0.088).

TECHNICAL SKILLS

Languages: Python, R, SQL, JavaScript, TypeScript
Statistics and Analytics: Statistical Modeling, Causal Inference (Difference-in-Differences, Event Study, Synthetic Control), Power Analysis, Econometrics (Panel Regression, Fixed Effects), Hypothesis Testing, Experimental Design, Statistical Significance Testing, Descriptive Statistics, Correlation Analysis, A/B Testing, Regression, Classification, Clustering, Forecasting (Prophet), Feature Engineering, Exploratory Data Analysis, Model Evaluation, Association Rule Mining, Data Visualization
Machine Learning and AI: Scikit-learn, XGBoost, SHAP, Random Forest, Decision Trees, PyTorch, TensorFlow, NLP, Computer Vision, LLMs, GenAI, LoRA Fine-tuning, RAG, LangChain, RAGAS, Hugging Face, Ollama
Data and Tools: Pandas, NumPy, Matplotlib, Power BI, Streamlit, PostgreSQL, MySQL, SQLite, Snowflake, Airflow, ETL, MLflow, Docker, Git, AWS, GCP, Azure

EDUCATION

University of Colorado, Boulder **Aug 2025 – May 2027**
Master of Science in Data Science Boulder, CO

Nirma University **Oct 2021 – May 2025**
Bachelor of Technology in Electronics and Communication, Minor in Computer Science Ahmedabad, India
Relevant Coursework: Machine Learning, Statistical Methods and Applications I and II, Data Mining, Data Economics